CHENGYUAN WANG (ERIC)

Computer Science Ph.D. Student | Formal Methods & Compilers

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EXPERIENCE

POSIX Shell Compiler Verification

Target venue: POPL 2028

Oct. 2023 – Present Hoboken, USA

- Develop a **Coq** model of core POSIX shell features, including pipelines and background commands, with process-management and filesystem semantics.
- Verify equivalence between sequential shell execution and programs automatically parallelized by PaSh.

Incrementalization System Verification

ATLAS Lab, Brown University

Jun. 2026 – Present Providence, USA

Target venue: PLDI 2027

- Formalize the execution semantics of a bolt-on incrementalization system.
- Establish an equivalence relation between incremental execution and standard shell execution under specified assumptions.

eBPF Verification in Lean

ATLAS Lab, Brown University

Jul. 2026 – Present Providence, USA

- Built a C-like DSL in Lean for writing eBPF programs, with support for eBPF helper functions and familiar C-style programming constructs.
- Project goal: integrate this work with the Linux kernel to support the eBPF verifier.

Safe Gradual Type Migration

Target venue: PLDI 2027

Sep. 2022 – Jun. 2024 Hoboken, USA

- Designed an inference-based algorithm for migrating untyped programs to gradually typed programs.
- Defined migration safety as introducing no new static errors, and proved this property for the proposed algorithm and existing migration approaches.

Static Policy Enforcement for a Functional Programming Language

Master's Thesis

Feb. 2022 – Jun. 2022 Gothenburg, Sweden

- Formalized the GDPR policy of purpose limitation.
- Designed an **information-flow control type system** to check purpose limitation statically and implemented it in Haskell.
- Formalized the type system in **Agda** and proved soundness and noninterference.

BaaS (Backend as a Service) Framework

Synyi Tech

Jun. 2021 – Aug. 2021 Shanghai, China

- Contributed to a Go framework for a medical information system, combining automatic CRUD API generation, policy-based authentication, and a TypeScript DSL for frontend interaction.

Compiler for a Fragment of Java

Individual Project

Mar. 2021 – Jun. 2021 Gothenburg, Sweden

- Implemented a lexer, parser, and **LLVM IR** code generator in Rust for a subset of Java.
- Implemented **LLVM optimization passes** for the generated code.

A Neural Rendering Framework for Free-Viewpoint Relighting

MARS Lab

Sep. 2019 – Feb. 2020 Shanghai, China

- Built a **relightable rendering model**. Used a GNN to extract 3D features from point clouds in the feature extractor. **Paper was accepted by CVPR 2020**.

R&D Intern

DGene Tech

Mar. 2019 – May. 2020 Shanghai, China

- Participated in a 3D face reconstruction project, responsible for building and compressing a neural network and deploying it on an Android phone.

EDUCATION

Ph.D., Computer Science

Stevens Institute of Technology

Sep 2022 – Present

Expected graduation: May 2028

Advisor: Michael Greenberg

M.Eng., Computer Science and Engineering

Chalmers University of Technology

Sep 2020 – Jun 2022

B.Eng., Computer Science and Engineering

Shanghai University

Aug 2016 – Jun 2020

SKILLS

- **Languages:** Rust, C/C++, Haskell, Go, Python
- **Proof Assistants:** Lean, Coq, Agda
- **Compiler Tools:** LLVM
- **Spoken Languages:** Mandarin, English

PUBLICATIONS

Accepted: CVPR 2020

A Neural Rendering Framework for Free-Viewpoint Relighting

Z. Chen, A. Chen, G. Zhang, **C. Wang**, Y. Ji, K. N. Kutulakos, J. Yu.

Paper

Target venue: POPL 2028

POSIX Shell Compiler Verification

Target venue: PLDI 2027

Safe Gradual Type Migration

Incrementalization System Verification

RESEARCH INTERESTS

- Programming languages, type systems, and effect handlers
- Compiler optimization and correctness
- Language-based security, information-flow control, and system verification